

Data Privacy: Midterm Solutions

1. **The Statistical Price of Local Privacy (20 pts)** A medical researcher wishes to estimate the fraction $p = \frac{1}{n} \sum_{i=1}^n x_i$ of individuals in a population of size n who carry a specific rare gene, where $x_i \in \{0, 1\}$ represents the true status of individual i . The study must satisfy ε -Differential Privacy.

We will analyze the statistical utility (measured by the variance of the error) in two different trust models: the **Central Model** and the **Local Model**.

- (a) **(5 pts)** Assume there is a completely trusted curator who collects the true bits $x_1, \dots, x_n$, computes the exact fraction p , and uses the Laplace Mechanism to release a private estimate $\hat{p}_{\text{central}}$.
 1. State the exact distribution of the noise that should be added to p .
 2. Calculate the exact variance of $\hat{p}_{\text{central}}$.
- (b) **(5 pts)** Now assume there is no trusted curator. Each user i must locally perturb their bit x_i before sending it to the researcher. They randomized their bit, and sending a bit $Y_i \in \{0, 1\}$ where:

$$\Pr[Y_i = 1 \mid x_i = 1] = \frac{e^\varepsilon}{1 + e^\varepsilon} \quad \text{and} \quad \Pr[Y_i = 1 \mid x_i = 0] = \frac{1}{1 + e^\varepsilon}$$

The researcher calculates the sum of the noisy reports $S = \sum_{i=1}^n Y_i$, and constructs the unbiased estimator $\hat{p}_{\text{local}}$ for the true fraction p :

$$\hat{p}_{\text{local}} = \frac{S}{n} \left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right) - \frac{1}{e^\varepsilon - 1}$$

Calculate the exact variance of $\hat{p}_{\text{local}}$.

- (c) **(10 pts)** Using the approximation $e^\varepsilon \approx 1 + \varepsilon$ for small ε , find the approximate asymptotic variances for both $\hat{p}_{\text{central}}$ and $\hat{p}_{\text{local}}$. If the researcher wants the Local DP estimator to achieve the *exact same variance* as the Central DP estimator for a fixed ε , how much larger must the dataset size n_{local} be compared to n_{central} ? Express n_{local} in terms of n_{central} .

Solution.

- (a) **1.** The query is the fraction $p = \frac{1}{n} \sum_{i=1}^n x_i$. Changing one person's bit x_i from 0 to 1 (or vice versa) changes the total sum by 1, so it changes the fraction p by exactly $\Delta = \frac{1}{n}$. Therefore, the noise added is drawn from $\text{Lap}\left(\frac{\Delta}{\varepsilon}\right) = \text{Lap}\left(\frac{1}{n\varepsilon}\right)$.
2. The variance of a Laplace distribution $\text{Lap}(b)$ is $2b^2$. Thus, the variance of the central estimator is:

$$\text{Var}(\hat{p}_{\text{central}}) = 2 \left(\frac{1}{n\varepsilon} \right)^2 = \frac{2}{n^2\varepsilon^2}$$

- (b) First, we find the variance of a single user's report Y_i . Since $Y_i \in \{0, 1\}$, it is a Bernoulli random variable. The variance of a Bernoulli variable with success probability q is $q(1 - q)$.

Let $q = \frac{e^\varepsilon}{1+e^\varepsilon}$. If $x_i = 1$, $Y_i \sim \text{Bernoulli}(q)$. Its variance is $q(1-q)$. If $x_i = 0$, $Y_i \sim \text{Bernoulli}(1-q)$. Its variance is $(1-q)(q) = q(1-q)$. Notice the variance is identical regardless of the underlying bit x_i :

$$\text{Var}(Y_i) = \left(\frac{e^\varepsilon}{1+e^\varepsilon} \right) \left(\frac{1}{1+e^\varepsilon} \right) = \frac{e^\varepsilon}{(1+e^\varepsilon)^2}$$

Because the Y_i reports are independent, the variance of the sum $S = \sum Y_i$ is simply n times the individual variance:

$$\text{Var}(S) = n \frac{e^\varepsilon}{(1+e^\varepsilon)^2}$$

Now we find the variance of the estimator $\hat{p}_{\text{local}}$. Using the variance property $\text{Var}(aX + c) = a^2 \text{Var}(X)$:

$$\text{Var}(\hat{p}_{\text{local}}) = \text{Var} \left(\frac{S}{n} \left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right) - \frac{1}{e^\varepsilon - 1} \right) = \frac{1}{n^2} \left(\frac{e^\varepsilon + 1}{e^\varepsilon - 1} \right)^2 \text{Var}(S)$$

Substitute $\text{Var}(S)$ into the equation:

$$\text{Var}(\hat{p}_{\text{local}}) = \frac{1}{n^2} \frac{(e^\varepsilon + 1)^2}{(e^\varepsilon - 1)^2} \cdot n \frac{e^\varepsilon}{(1+e^\varepsilon)^2}$$

The $(e^\varepsilon + 1)^2$ terms cancel out, leaving:

$$\text{Var}(\hat{p}_{\text{local}}) = \frac{e^\varepsilon}{n(e^\varepsilon - 1)^2}$$

(c) Using the approximation $e^\varepsilon \approx 1 + \varepsilon$ for small ε :

- $\hat{p}_{\text{central}}$ variance is already independent of this approximation: $\approx \mathcal{O}\left(\frac{1}{n^2 \varepsilon^2}\right)$
- $\hat{p}_{\text{local}}$ variance simplifies to: $\frac{1+\varepsilon}{n(1+\varepsilon-1)^2} = \frac{1+\varepsilon}{n\varepsilon^2} \approx \frac{1}{n\varepsilon^2}$

To find how much more data is required for the Local model to match the Central model's accuracy, we equate the variances:

$$\text{Var}(\hat{p}_{\text{local}}) = \text{Var}(\hat{p}_{\text{central}})$$

$$\frac{1}{n_{\text{local}} \varepsilon^2} = \frac{2}{n_{\text{central}}^2 \varepsilon^2}$$

Solving for n_{local} :

$$n_{\text{local}} = \frac{1}{2} n_{\text{central}}^2$$

2. **Hypothesis Testing and Approximate Differential Privacy (40 pts)** Let D and D' be adjacent datasets. An adversary observes the output of a mechanism $\mathcal{M}$ and conducts a hypothesis test to guess the true dataset:

$$H_0: \text{The dataset is } D, \quad H_1: \text{The dataset is } D'.$$

Using a rejection region S , the adversary declares H_1 if the output falls in S , and H_0 otherwise. We formally define the error probabilities:

- **False Positive (q):** $\Pr(\text{declare } H_1 \mid H_0 \text{ is true}) = \Pr[\mathcal{M}(D) \in S]$
- **False Negative (p):** $\Pr(\text{declare } H_0 \mid H_1 \text{ is true}) = \Pr[\mathcal{M}(D') \notin S]$

For pure ϵ -DP, the trade-off between these errors is strictly bounded by $q + pe^\epsilon \geq 1$ and $p + qe^\epsilon \geq 1$. We now extend this to approximate (ϵ, δ) -DP.

- (8 pts) Suppose $\mathcal{M}$ satisfies (ϵ, δ) -DP. Using the definition of (ϵ, δ) -DP, derive the new lower bounds for $q + pe^\epsilon$ and $p + qe^\epsilon$ in terms of ϵ and δ .
- (8 pts) Suppose a mechanism satisfies $(0, 1)$ -DP. Plug $\epsilon = 0$ and $\delta = 1$ into your bounds. Briefly explain what this implies about the adversary's ability to achieve $p = 0$ and $q = 0$ simultaneously, and why this confirms $\delta = 1$ provides no privacy.
- (8 pts) Consider a mechanism $\mathcal{M}$ outputting values in $\{0, 1, 2\}$ with the following probabilities:

$$\begin{aligned}\Pr[\mathcal{M}(D) = z] &= \{0.5, 0.5, 0\} \quad \text{for } z \in \{0, 1, 2\} \\ \Pr[\mathcal{M}(D') = z] &= \{0, 0.5, 0.5\} \quad \text{for } z \in \{0, 1, 2\}\end{aligned}$$

Assume the adversary uses the rejection region $S = \{1, 2\}$. Calculate the exact False Positive rate q and False Negative rate p for this specific test.

- (8 pts) Prove that the mechanism defined in part (c) cannot satisfy pure ϵ -DP for any finite ϵ .
- (8 pts) Find the minimum δ such that the mechanism from part (c) satisfies $(\ln 2, \delta)$ -DP. Verify that your p and q values mathematically satisfy the bounds you derived in part (a).

Solution.

- By the definition of (ϵ, δ) -DP, for any subset of outputs S :

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$$

Substituting our error probabilities $q = \Pr[\mathcal{M}(D) \in S]$ and $(1-p) = \Pr[\mathcal{M}(D') \in S]$:

$$q \leq e^\epsilon(1-p) + \delta$$

Rearranging the terms yields the first bound:

$$q + pe^\epsilon \geq 1 - \delta$$

By symmetry, applying the DP definition from D' to D for the event S^c :

$$\Pr[\mathcal{M}(D') \in S^c] \leq e^\epsilon \Pr[\mathcal{M}(D) \in S^c] + \delta$$

Substituting $p = \Pr[\mathcal{M}(D') \in S^c]$ and $(1-q) = \Pr[\mathcal{M}(D) \in S^c]$:

$$p \leq e^\epsilon(1-q) + \delta \implies p + qe^\epsilon \geq 1 - \delta$$

- (b) Substituting $\epsilon = 0$ and $\delta = 1$ into the bound $q + pe^\epsilon \geq 1 - \delta$ gives:

$$q + p(1) \geq 1 - 1 \implies q + p \geq 0$$

Because probabilities are always non-negative, $q + p \geq 0$ is a mathematical triviality that places absolutely no restriction on the adversary. An adversary could design a perfect test where $q = 0$ and $p = 0$ (100% accuracy in distinguishing D from D'), and it would still satisfy this inequality. Thus, $\delta = 1$ offers zero privacy protection.

- (c) The rejection region is $S = \{1, 2\}$.

$$\begin{aligned} q &= \Pr[\mathcal{M}(D) \in \{1, 2\}] = 0.5 + 0 = 0.5 \\ p &= \Pr[\mathcal{M}(D') \notin \{1, 2\}] = \Pr[\mathcal{M}(D') = 0] = 0 \end{aligned}$$

Thus, $p = 0$ and $q = 0.5$.

- (d) For pure DP to hold, the ratio of probabilities for any event must be bounded by e^ϵ . Consider the output $z = 2$:

$$\Pr[\mathcal{M}(D') = 2] \leq e^\epsilon \Pr[\mathcal{M}(D) = 2] \implies 0.5 \leq e^\epsilon \cdot 0$$

This requires $0.5 \leq 0$, which is mathematically impossible for any finite ϵ . Thus, it cannot satisfy pure ϵ -DP.

- (e) To find the minimum δ for $\epsilon = \ln 2$ (which means $e^\epsilon = 2$), we check the DP condition $\Pr[D'] \leq 2\Pr[D] + \delta$ and $\Pr[D] \leq 2\Pr[D'] + \delta$ across all subsets T :

- For $T = \{0\}$: $\Pr[\mathcal{M}(D) = 0] = 0.5$ and $\Pr[\mathcal{M}(D') = 0] = 0$.
 $0.5 \leq 2(0) + \delta \implies \delta \geq 0.5$.
- For $T = \{2\}$: $\Pr[\mathcal{M}(D') = 2] = 0.5$ and $\Pr[\mathcal{M}(D) = 2] = 0$.
 $0.5 \leq 2(0) + \delta \implies \delta \geq 0.5$.

Other subsets yield less strict bounds (e.g., $T = \{1, 2\}$ gives $1 \leq 2(0.5) + \delta \implies \delta \geq 0$). Thus, the minimum δ is 0.5.

Now, verify the (p, q) bounds using $p = 0$, $q = 0.5$, $e^\epsilon = 2$, and $\delta = 0.5$:

$$\begin{aligned} q + pe^\epsilon &= 0.5 + 0(2) = 0.5 \\ 1 - \delta &= 1 - 0.5 = 0.5 \end{aligned}$$

Since $0.5 \geq 0.5$, the bound $q + pe^\epsilon \geq 1 - \delta$ holds perfectly.

3. **Tightness of the Exponential Mechanism (40 pts)** Consider an election with d candidates. The scoring function $q(D, y)$ is the number of votes candidate y receives in dataset D . The sensitivity is $\Delta = 1$.

To understand the theoretical limits of differential privacy, we will construct a family of extreme, “easy” datasets: $D^{(1)}, D^{(2)}, \dots, D^{(d)}$. In a specific dataset $D^{(j)}$, exactly one candidate j receives all n votes, and all other $d - 1$ candidates receive 0 votes. For this entire problem, assume the number of voters n is fixed such that $n = \frac{\ln(d)}{2\epsilon}$.

- (a) **(10 pts)** Suppose we run the standard Exponential Mechanism $\mathcal{A}_{EM}$ on the dataset $D^{(j)}$. Write down the unnormalized weight w_j of the winning candidate j , and the weight w_k of any losing candidate k . (Simplify your answers using the given value of n).

- (b) **(10 pts)** Using your weights from part (a), show that $\mathcal{A}_{EM}$ outputs a candidate *other* than j with at least some constant probability, independent of d (for $d \geq 2$).
- (c) **(10 pts)** Now we want to show that **no** ε -DP algorithm can do much better. Let $\mathcal{A}$ be an arbitrary ε -DP algorithm. Let $x_\emptyset$ be the completely empty dataset where all candidates have 0 votes.
- Using the group privacy property of DP, find an upper bound for the probability $\Pr[\mathcal{A}(D^{(j)}) = j]$ strictly in terms of d and $\Pr[\mathcal{A}(x_\emptyset) = j]$.
- (d) **(10 pts)** Finally, let's evaluate the average accuracy of $\mathcal{A}$ across all d extreme datasets. Equivalently, imagine an adversary chooses a true winning candidate J uniformly at random from $[d]$, and feeds the corresponding dataset $D^{(J)}$ to your algorithm. Using your bound from part (c), prove that the algorithm fails to output J with at least a constant probability.
- Hint: Since the algorithm must output some candidate when run on $x_\emptyset$, the sum of probabilities $\sum_{j=1}^d \Pr[\mathcal{A}(x_\emptyset) = j]$ must equal 1.*

Solution.

- (a) The Exponential Mechanism outputs a candidate y with probability proportional to $\exp\left(\frac{\varepsilon q(D, y)}{2\Delta}\right)$. With $\Delta = 1$:

- Weight of the winning candidate j :

$$w_j = \exp\left(\frac{\varepsilon n}{2}\right) = \exp\left(\frac{\varepsilon}{2} \cdot \frac{\ln(d)}{2\varepsilon}\right) = \exp\left(\frac{\ln(d)}{4}\right) = d^{1/4}$$

- Weight of any other candidate $k \neq j$:

$$w_k = \exp\left(\frac{\varepsilon \cdot 0}{2}\right) = \exp(0) = 1$$

- (b) The probability of outputting a candidate other than j is the sum of the weights of the $d - 1$ losing candidates divided by the total partition weight:

$$\Pr[\mathcal{A}_{EM}(D^{(j)}) \neq j] = \frac{d - 1}{d^{1/4} + d - 1}$$

To show this is bounded below by a constant independent of d , we evaluate the worst-case minimum at $d = 2$:

$$\Pr[\mathcal{A}_{EM}(D^{(j)}) \neq j] \geq \frac{1}{2^{1/4} + 1} \approx 0.456$$

Thus, the algorithm fails with at least a $\sim 45.6\%$ probability, independent of d .

- (c) The L_1 distance (number of differing votes) between $D^{(j)}$ and $x_\emptyset$ is exactly $n = \frac{\ln(d)}{2\varepsilon}$. By the group privacy property, for any specific candidate j :

$$\begin{aligned} \Pr[\mathcal{A}(D^{(j)}) = j] &\leq e^{\varepsilon n} \Pr[\mathcal{A}(x_\emptyset) = j] \\ &= \exp\left(\varepsilon \cdot \frac{\ln(d)}{2\varepsilon}\right) \Pr[\mathcal{A}(x_\emptyset) = j] \\ &= \sqrt{d} \Pr[\mathcal{A}(x_\emptyset) = j] \end{aligned}$$

- (d) We want to find the expected probability of success when J is chosen uniformly at random:

$$\mathbb{E}_J [\Pr[\mathcal{A}(D^{(J)}) = J]] = \frac{1}{d} \sum_{j=1}^d \Pr[\mathcal{A}(D^{(j)}) = j]$$

Substituting the group privacy bound from part (c) into the sum:

$$\mathbb{E}_J [\Pr[\mathcal{A}(D^{(J)}) = J]] \leq \frac{1}{d} \sum_{j=1}^d \sqrt{d} \Pr[\mathcal{A}(x_\emptyset) = j] = \frac{1}{\sqrt{d}} \sum_{j=1}^d \Pr[\mathcal{A}(x_\emptyset) = j]$$

Since the algorithm must output exactly one candidate, $\sum_{j=1}^d \Pr[\mathcal{A}(x_\emptyset) = j] = 1$. Therefore:

$$\mathbb{E}_J [\Pr[\mathcal{A}(D^{(J)}) = J]] \leq \frac{1}{\sqrt{d}} \cdot 1 = \frac{1}{\sqrt{d}}$$

The probability that the algorithm *fails* to identify J is the complement:

$$\Pr[\mathcal{A}(D^{(J)}) \neq J] = 1 - \Pr[\mathcal{A}(D^{(J)}) = J] \geq 1 - \frac{1}{\sqrt{d}}$$

For any $d \geq 2$, this failure probability is at least $1 - \frac{1}{\sqrt{2}} \approx 0.293$. Thus, any ε -DP algorithm fails with at least a strictly positive constant probability.